

PROJECT REPORT

Cisco Virtual Internship 2026

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Abstract

This project presents the design and implementation of a small-office/branch network for SmartBranch 360. The network uses one router, two switches, a wireless access point, a server, an Internet/ISP connection and multiple endpoints. The LAN is segmented into Employee, Guest, Server and Management VLANs. Router-on-a-stick provides inter-VLAN routing, DHCP provides automatic addressing, DNS supports name resolution, NAT overload provides Internet address translation, ACLs enforce traffic restrictions, and SSH provides secure management. A Python assurance tool validates important Cisco configuration elements and reports common problems.

The report is based on the given problem statement and the router configuration designed for the topology

Problem Statement

The supplied SmartBranch 360 problem statement asks for a small office network with wired employees,

guest Wi-Fi, an internal server, Internet access and secure management of network devices. It specifically

requires separate Employee, Guest, Server and Management VLANs, DHCP per VLAN, DNS reachability,

Internet NAT/routing, guest isolation from the server VLAN, and SSH management only from the

management VLAN.

The problem statement also requires a troubleshooting workflow and a small Python tool that reads a plan or

show-command output and reports likely configuration problems such as a missing VLAN, incorrect gateway,

or an ACL that blocks required traffic.

1. Project Requirements

Area	Requirement
Topology	1 router, 2 switches, 1 wireless AP, 1 server, Internet/cloud and 8 or more endpoints.
VLANs	Employee VLAN 10, Guest VLAN 20, Server VLAN 30 and Management VLAN 99.
Services	DHCP per VLAN, DNS reachability, NAT for Internet access and inter-VLAN routing.
Security	Guest isolation and secure SSH-based management.
Faults	Wrong gateway, missing VLAN on trunk, bad DHCP, ACL blocking DNS, NAT issue.
Assurance	Python tool validates Packet Tracer design/show output and reports configuration problems.

2. Objectives

- Segment the branch network using VLANs.
- Configure router-on-a-stick inter-VLAN routing.
- Provide DHCP and DNS services.
- Provide Internet access through NAT overload.
- Restrict Guest traffic with ACLs.
- Provide SSH management from the Management network.
- Create a Python configuration-assurance/troubleshooting tool.
- Verify the network using show commands, ping, traceroute and Simulation Mode.

3. Network Architecture

The branch router provides the Layer-3 gateway for all four VLANs. G0/0/0 is the LAN-side trunk and uses subinterfaces for VLANs 10, 20, 30 and 99. G0/0/1 is the WAN interface toward the ISP. The logical traffic path is:

Endpoint → Access Switch → Trunk → R1 VLAN subinterface → R1 WAN → ISP → Internet

3.1 VLAN/IP Plan

VLAN	Name	Network	Gateway	Purpose
10	Employee	10.10.10.0/24	10.10.10.1	Employee devices
20	Guest	10.10.20.0/24	10.10.20.1	Guest devices
30	Server	10.10.30.0/24	10.10.30.1	Server/DNS services
99	Management	10.10.99.0/24	10.10.99.1	Network management

WAN: R1 G0/0/1 = 203.0.113.2/24; ISP next hop = 203.0.113.1.

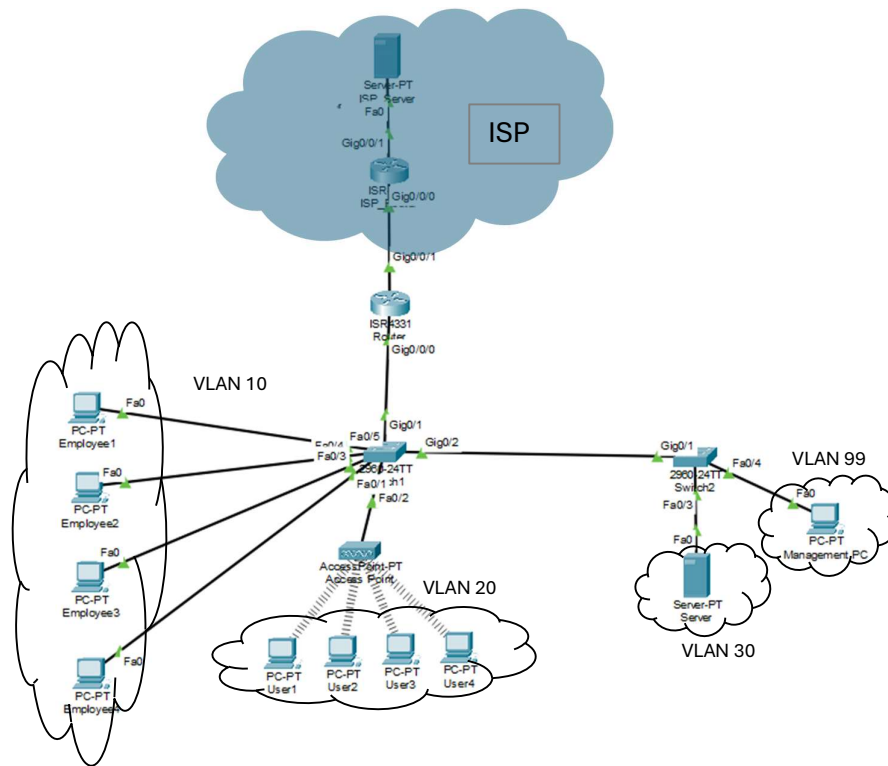


Fig .1 Network Topology

4. Components and Roles

Component	Role
R1 Router	Inter-VLAN routing, DHCP, NAT, ACLs, default route and SSH.
Switch 1	Access switching and trunk connectivity.
Switch 2	Additional access switching and trunk extension.
Wireless AP	Wireless endpoint connectivity.
Server	Server VLAN host and DNS/internal services.
ISP Router	WAN next hop/simulated Internet.
Endpoints	Employee, Guest and Management testing clients.

5. Configuration

5.1 Router-on-a-Stick

I. VAN10 Employee Network

```
interface GigabitEthernet0/0/0.10
encapsulation dot1Q 10
ip address 10.10.10.1 255.255.255.0
ip nat inside
```

II. VLAN20 Guest Network

```
interface GigabitEthernet0/0/0.20
encapsulation dot1Q 20
ip address 10.10.20.1 255.255.255.0
ip nat inside
```

III. VLAN30 Server Network

```
interface GigabitEthernet0/0/0.30
encapsulation dot1Q 30
ip address 10.10.30.1 255.255.255.0
ip nat inside
```

IV. VLAN 99 Management Network

```
interface GigabitEthernet0/0/0.99
encapsulation dot1Q 99
ip address 10.10.99.1 255.255.255.0
ip nat inside
```

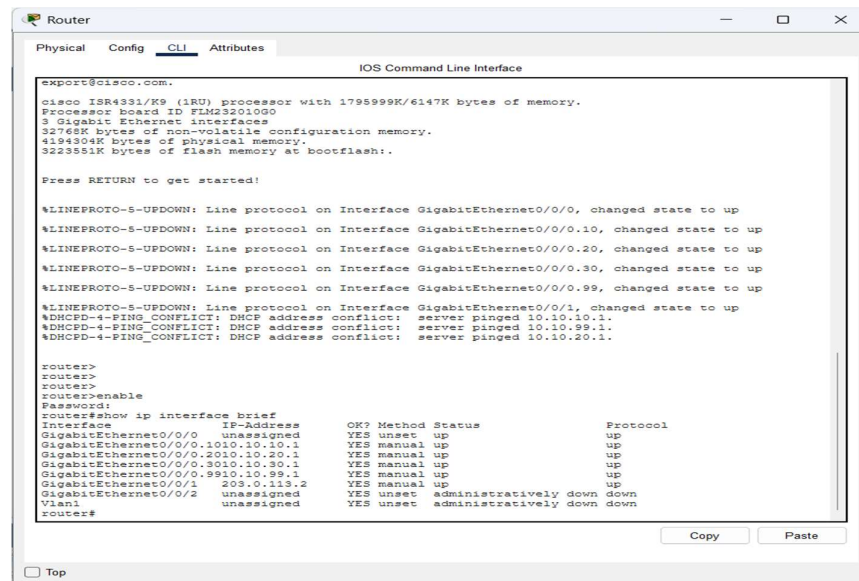


Fig.2 Network configuration

5.2 DHCP

Four DHCP pools are configured for Employee, Guest, Server and Management. Each pool provides the corresponding subnet and default gateway.

I. Employee

```
ip dhcp pool employee
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 10.10.30.10
```

II. Guests

```
ip dhcp pool guests
network 10.10.20.0 255.255.255.0
default-router 10.10.20.1
dns-server 10.10.30.10
```

III. Server

```
ip dhcp pool server
network 10.10.30.0 255.255.255.0
default-router 10.10.30.1
dns-server 10.10.30.10
```

IV. Management

```
ip dhcp pool management
network 10.10.99.0 255.255.255.0
default-router 10.10.99.1
dns-server 10.10.30.10
```

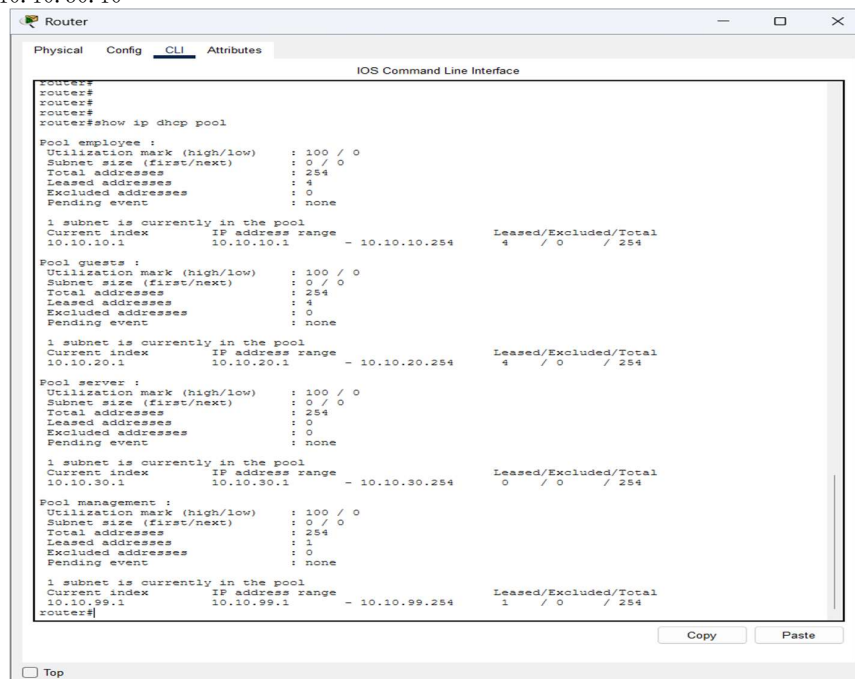


Fig.3 DHCP configuration

5.3 NAT/WAN

The project was extended beyond the internal VLANs to an ISP/WAN path. The intended flow is

Employee/Guest/other internal VLAN → R1 → WAN/ISP → Internet/Internet server.
Traceroute was used as

a diagnostic tool to identify hops.

A traceroute to 10.10.10.3 showed that a destination inside the same /24 subnet is normally reached directly

through Layer-2/ARP rather than by routing through the default gateway. For an actual Internet destination

outside 10.10.10.0/24, the first routed hop should normally be the VLAN gateway 10.10.10.1.

```
interface GigabitEthernet0/0/1
ip address 203.0.113.2 255.255.255.0
ip nat outside

ip nat inside source list 1 interface GigabitEthernet0/0/1 overload
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

```
router#show running-config | include ip nat
ip nat inside
ip nat inside
ip nat inside
ip nat inside
ip nat outside
ip nat inside source list 1 interface GigabitEthernet0/0/1 overload
router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/0/1
Inside Interfaces: GigabitEthernet0/0/0.10 , GigabitEthernet0/0/0.20 , GigabitEthernet0/0/0.30 ,
GigabitEthernet0/0/0.99
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
router#
```

Fig.4 Ip nat configuration

5.4 NAT ACL

```
access-list 1 permit 10.10.10.0 0.0.0.255
access-list 1 permit 10.10.20.0 0.0.0.255
access-list 1 permit 10.10.30.0 0.0.0.255
access-list 1 permit 10.10.99.0 0.0.0.255
```

5.5 Switching & Trunk

Access ports are assigned to the appropriate VLAN, while uplinks between switches/router are configured as trunks. During troubleshooting, the allowed VLAN list and native VLAN were important because a trunk must

carry the VLANs required by downstream endpoints.

A native-VLAN mismatch was observed during configuration. This was a useful fault scenario: one side of a

trunk was treating a port as native VLAN 1 while the other side had a different native/access expectation. The

mismatch was identified by the Packet Tracer/CDP warning and corrected by aligning trunk configuration.

```
Switch(config)#interface gig0/2
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,99
Switch(config-if)#no shutdown
```


6. Security

6.1 VLAN Segmentation

- VLAN 10: employee endpoints.
- VLAN 20: guest wireless endpoints.
- VLAN 30: servers and internal services.
- VLAN 99: management hosts and management addresses.

6.2 Guest Isolation

```
ip access-list extended GuestSecurity
deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.99.0 0.0.0.255
permit ip any any
```

This prevents Guest clients from accessing the Server, Employee and Management networks while allowing other permitted destinations.

6.3 Employee and Management

The current EmployeeSecurity ACL blocks Employee-to-Management traffic and permits other traffic. The current ManagementSecurity ACL is permissive (permit ip any any), so it should not be described as an isolation ACL.

6.3 Secure Management with SSH + Radius

The project uses centralized RADIUS authentication for administrative logins. The RADIUS server was configured with authentication port 1645 and a user account named networkadmin. R1, SW1 and SW2 were registered as RADIUS clients on the server. The intended security rule is that valid credentials are not sufficient by themselves: SSH management is permitted only when the source host is in VLAN 99.

1.SSH

```
ip ssh version 2
username admin privilege 15 secret <configured-secret>
ip access-list standard SSH_Management
permit host 10.10.99.10
line vty 0 4
access-class SSH_Management in
login local
transport input ssh
```

2.Radius

The Packet Tracer router accepted the newer **radius server** configuration style, with the RADIUS server address set to **10.10.30.10** and the configured shared key. The RADIUS server is placed in **VLAN 30 (Server VLAN)**, while the Management PC is located in **VLAN 99 (Management VLAN)** and accesses the router through **SSH**. Since the simulator IOS did not support the **ip radius source-interface** command, the RADIUS client/source-IP behaviour was verified through connectivity and SSH authentication testing.

```
radius server RADIUS1
address ipv4 10.10.30.10 auth-port 1645
key radius123
radius server 10.10.30.10
address ipv4 10.10.30.10 auth-port 1645
key radius123
```

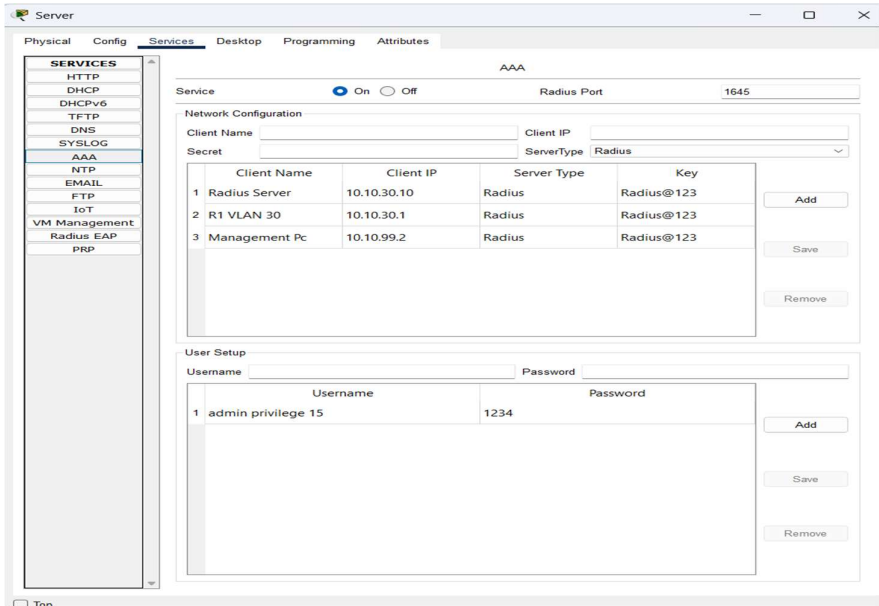


Fig.5 radius configuration

7. Verification

Test	Command/Action	Expected Result
Interfaces	show ip interface brief	Required interfaces are up/up.
Routing	show ip route	Connected VLAN routes and default route exist.
DHCP	show ip dhcp binding	Clients have correct leases.

Gateway	ping own gateway	Successful reply.
ISP	ping 203.0.113.1	WAN next-hop reachable.
NAT	show ip nat translations	Translations appear after Internet-bound traffic.
NAT stats	show ip nat statistics	NAT activity is reported.
ACL	show access-lists	Match counters change with traffic.
Trunk	show interfaces trunk	Required VLANs are carried.
Traffic	Packet Tracer Simulation Mode	Packets can be followed hop-by-hop.

For Internet verification, test an endpoint in each relevant VLAN: ipconfig → ping its gateway → ping the ISP next hop → ping a reachable Internet/test server. Then inspect NAT translations and statistics.

Router's running configuration Output:

```

router#show r
Building configuration...
Current configuration : 2590 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname router
!
enable secret 5 $1$mERr$4dpRATigxQacPVKOCfNV4/
enable password 1234
!
ip dhcp pool employee
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 10.10.30.10
ip dhcp pool guests
network 10.10.20.0 255.255.255.0
default-router 10.10.20.1
dns-server 10.10.30.10
ip dhcp pool server
network 10.10.30.0 255.255.255.0
default-router 10.10.30.1
dns-server 10.10.30.10
ip dhcp pool management
network 10.10.99.0 255.255.255.0
default-router 10.10.99.1
dns-server 10.10.30.10
!
ip cef

```

```
no ipv6 cef
!
username admin privilege 15 secret 5 $1$mERr$gjNBRlu.Is9z7oaYvbvh//
!
ip ssh version 2
ip domain-name SmartBranch360.local
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0/0.10
encapsulation dot1Q 10
ip address 10.10.10.1 255.255.255.0
ip nat inside
ip access-group EmployeeSecurity in
!
interface GigabitEthernet0/0/0.20
encapsulation dot1Q 20
ip address 10.10.20.1 255.255.255.0
ip nat inside
ip access-group GuestSecurity in
!
interface GigabitEthernet0/0/0.30
encapsulation dot1Q 30
ip address 10.10.30.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/0.99
encapsulation dot1Q 99
ip address 10.10.99.1 255.255.255.0
ip nat inside
ip access-group ManagementSecurity in
!
interface GigabitEthernet0/0/1
ip address 203.0.113.2 255.255.255.0
ip nat outside
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
mac-address 0002.4a67.1501
no ip address
shutdown
!
ip nat inside source list 1 interface GigabitEthernet0/0/1 overload
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
ip flow-export version 9
```

```

!
ip access-list extended GuestSecurity
deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.99.0 0.0.0.255
permit ip any any
ip access-list extended EmployeeSecurity
deny ip 10.10.10.0 0.0.0.255 10.10.99.0 0.0.0.255
permit ip any any
ip access-list standard SSH_Management
permit host 10.10.99.10
access-list 1 permit 10.10.10.0 0.0.0.255
access-list 1 permit 10.10.20.0 0.0.0.255
access-list 1 permit 10.10.30.0 0.0.0.255
access-list 1 permit 10.10.99.0 0.0.0.255
!
line con 0
!
line aux 0
!
line vty 0 4
access-class SSH_Management in
login local
transport input ssh
!
end

```

8. Troubleshooting Fault Scenarios

Fault	Symptom	Checks	Fix
Wrong gateway	IP obtained but routing fails	ipconfig; ping gateway	Correct DHCP/host gateway.
Missing VLAN on trunk	One VLAN fails to reach gateway	show interfaces trunk	Create/allow VLAN on required trunk.
Bad DHCP	No/incorrect IP settings	show ip dhcp binding; ipconfig	Correct DHCP pool.
ACL blocks DNS	Names fail although IP may work	show access-lists; DNS test	Permit required DNS traffic.
NAT issue	Internet access fails	show ip nat translations; show ip route	Correct NAT roles, ACL, overload and route.

9. Python Assurance Tool

The Python tool is an assurance layer rather than a direct Packet Tracer configuration mechanism. The user pastes the router's show running-config output, ends input with END, and the script checks VLAN subinterfaces, DHCP pools, NAT/WAN settings, Guest ACL restrictions and SSH management.

Workflow:

- Configure Packet Tracer.
- Run show running-config on R1.
- Run the Python checker.
- Paste the configuration and type END.
- Review OK/ERROR findings.
- Fix the Cisco configuration.
- Repeat validation and perform connectivity tests.
- Project Python file: smartbranch360.py

Example:

```
[OK] Pool management gateway = 10.10.99.1
=====
NAT CHECK
=====
[ERROR] NAT overload is missing or uses the wrong interface.
=====
WAN / DEFAULT ROUTE CHECK
=====
[ERROR] Default route to 203.0.113.1 missing.
[ERROR] WAN IP 203.0.113.2 missing.
=====
ACL SECURITY CHECK
=====
[ERROR] GuestSecurity ACL is missing.
[ERROR] EmployeeSecurity ACL is missing.
=====
SSH MANAGEMENT CHECK
=====
[ERROR] SSH version 2 missing.
[ERROR] Domain name missing.
[ERROR] Administrator account missing.
[ERROR] VTY is not configured for SSH.
[ERROR] Local authentication missing.
[ERROR] SSH Management ACL missing.
[WARN] Management PC 10.10.99.3 not found in SSH configuration.
=====
FINAL SUMMARY
=====
RESULT: 7 major configuration item(s) may be missing.
```

Fig.6 Demonstration of the output of the python assurance tool

10. Expected Results

Employee clients receive 10.10.10.x addresses.

Guest clients receive 10.10.20.x addresses and cannot access protected internal VLANs.

Server VLAN uses 10.10.30.0/24 and provides the required server/DNS role.

Management uses 10.10.99.0/24 for administrative access.

All four internal networks are eligible for NAT.

Internet-bound traffic creates NAT translations when a reachable external destination is contacted.

SSH VTY access is restricted to the permitted management source.

Troubleshooting commands provide evidence of routing, DHCP, NAT, ACL and trunk operation.

11. Limitations

The supplied router configuration does not contain every physical Packet Tracer detail. Exact switch ports, AP settings, server application settings and the Internet test-server address must be verified in the .pkt topology. The current ManagementSecurity ACL is permissive, while SSH access is restricted separately by SSH_Management.

12. Conclusion

SmartBranch 360 demonstrates a segmented branch network using VLANs, router-on-a-stick routing, DHCP, DNS, NAT, ACLs and SSH. The design separates Employee, Guest, Server and Management traffic and provides a repeatable troubleshooting workflow. The Python assurance tool adds configuration validation to the Packet Tracer implementation and helps identify common faults before final verification.

Appendix – Useful Commands

```
show running-config
show ip interface brief
show ip route
show ip dhcp binding
show ip nat translations
show ip nat statistics
show access-lists
show interfaces trunk
ping <destination>
traceroute <destination>
```

***router config password:1234**